

DISPERSION OF LIGHT

In this chapter, we will study about some phenomena of light which are equally important as the transmission and reflection. Can you infer the phenomena, evident in the following pictures? **Hint:** there is one property of light that causes the following three phenomena.

In this Chapter you will learn about:

- Refraction
- Refraction in different mediums (glass and water)
- Laws of refraction and refractive index
- Real and apparent depth
- Critical angle (glass and water)
- Total internal reflection
- Applications: reflecting prisms, the periscope, mirages, fish eye view)
- Dispersion of light (spectrum and rainbow formation)
- Colours of light (primary and secondary colours)
- Colours of objects

All the students will be able to:

- ✓ Explain refraction of light and its causes
- ✓ Discuss the effects of refraction with examples
- ✓ List the colours of light using a prism
- ✓ Describe the dispersion of light by using a prism
- ✓ Identify different uses of light of different colours at home, school and country and explain the relationship of choice of colour to their purpose
- ✓ Define Spectrum of light
- ✓ Identify primary colours and show how they are combined to form secondary colours
- ✓ Identify a device in their surroundings that uses different combinations of colours
- ✓ Demonstrate how spinning of a rainbow results in the appearance of white disc
- ✓ Explain why an opaque or non-luminous object appears to be of certain colour



Figure 9.1: Is it a broken pencil?



Figure 9.2: Why does rainbow appear on Sky?

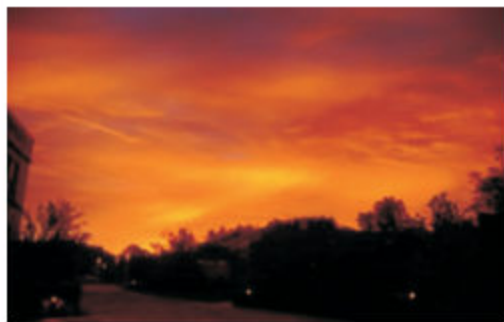


Figure 9.3: Why do the multi-coloured waves appear on Sky on Sunset?

REFRACTION OF LIGHT

✓ Explain refraction of light and its causes.

Change in the speed and direction of light rays due to change of medium is called refraction of light. When a light ray travels from a rare medium into denser medium it is refracted towards normal but when a ray of light enters into a rare medium from a denser medium it is refracted away from normal. The properties of a medium influence the behaviour of light. For instance, the density of a medium, changes both the speed and direction of light rays. The light travels fastest in the vacuum; while its speed is faster in the air as compared to the denser media such as water and glass. Moreover, the direction of light rays also changes on entering a denser medium.

Activity 9.1: Explore the refraction of light and its causes.

What I need:

Drawing board, sky blue drawing paper, a rectangular transparent glass slab, thumb tags, a torch, scissors, scotch tape and a square shaped cardboard as shown in 9.4 (a).

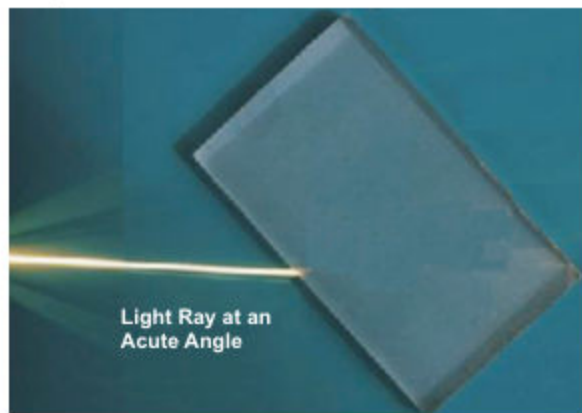


Figure 9.4 (a): Cardboard with hole

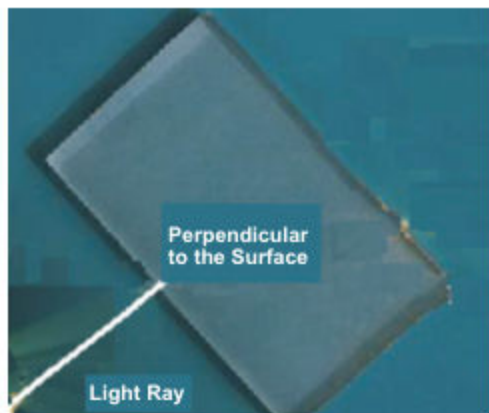


Figure 9.4 (b): Light ray at $30^\circ/60^\circ$ angle and perpendicular to the surface of glass slab

What to do:

1. Make a fine hole in the centre of the cardboard as shown in figure 9.4 (a).
2. Place a drawing board on the table.
3. Use thumb tags to fix the drawing paper on the board.
4. Fix the cardboard on a corner of the drawing board with the help of scotch tape.

- Place the glass slab in front of the cardboard.
- Lit torch in a way that light passes through the hole and enters the slab at (i) 30° and, (ii) 60° .
- Use ruler and trace the path ray of light on paper.
- Note the angles of the light ray before entering and after passing through the glass slab [(as shown in figure 9.4 (c))].
- Repeat the process with an adjustment of the position of the slab in a way that the light ray falls perpendicular on its surface.
- Use ruler and trace the ray of light on paper.

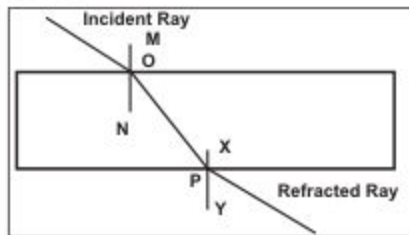


Figure 9.4 (c): Ray diagram of $30^\circ/60^\circ$

What I observe:

For Angle 30°	For Angle 60°	Perpendicular to glass's surface
Angle, incident ray make with normal line MON =	Angle, incident ray make with normal line MON =	Angle, incident ray make with normal line MON =
Angle refracted ray make with normal line XPY =	Angle refracted ray make with normal line XPY =	Angle refracted ray make with normal line XPY =
Mention the difference (if any) in the angle of incident ray and the angle of refracted ray =	Mention the difference (if any) in the angle of incident ray and the angle of refracted ray =	Mention the difference (if any) in the angle of incident ray and the angle of refracted ray =

Activity Questions:

- What is the direction of light ray:
 - When it falls at the acute angle (such as $30^\circ/60^\circ$)?
 - When it falls perpendicular to the surface of glass slab?
- What is the relation between the angle of the incident ray and the angle of refracted ray?
- Does change in the speed of light on entering the denser medium result into refraction?
- Does change in the angle of incident ray result into refraction irrespective of the speed of light?

Teacher Note: This is an individual activity to demonstrate the refraction of light. Ask students to follow the instructions carefully.

To sum up, when light passes from rare medium (i.e air) to the denser medium (i.e water) it bends towards normal and when light passes from denser medium (i.e water) to the rare medium (i.e air) it bends away from the normal (an imaginary line that can be drawn perpendicular to the surface of the medium such as glass or water). This bending of light is named as the refraction of light. If a light ray falls perpendicular to the surface of a denser medium such as glass and water, it changes its speed, not the direction (figure 9.4 (d)).

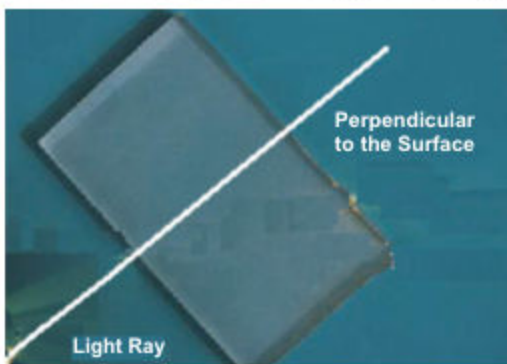


Figure 9.4 (d): No change in the direction of light

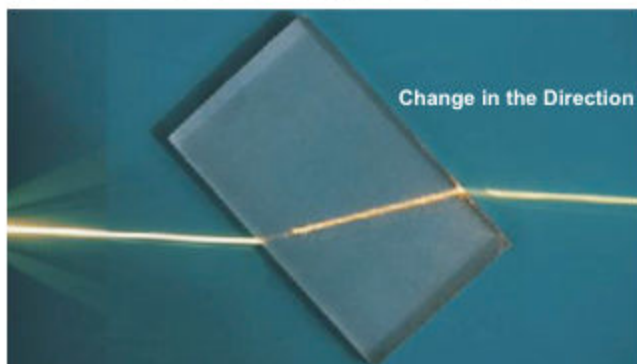


Figure 9.4 (e): Bending of light

Laws of Refraction and Refractive Index:

There are two laws of refraction.

- The incident ray, refracted ray and the normal at the point of incidence, all lie in the same plane.
- The ratio between the speed of light in vacuum and its speed in any other medium is constant. This ratio is known as the 'Refractive Index' of that particular medium.

DO YOU KNOW?

Laws of refraction are also known the Snell's Laws, named after the scientist Willebrord Snell who discovered these laws in 1621.

Refractive Index:

The refractive index is symbolized as 'n'. It can be calculated by the following formula:

$$\text{Refractive Index (n) of a medium} = \frac{\text{Speed of light in vacuum (c)}}{\text{Speed of light in the medium (v)}}$$

It has no unit of measurement because it is a ratio. Table 9.1 shows the refractive indices of different media.

Medium	Refractive Index	Medium	Refractive Index
Vacuum	1.000	Crown Glass	1.52
Air	1.0003	Diamond	2.42
Water	1.33	Amber glass	1.55

Table 9.1: Refractive indices of different media

REFRACTION IN DIFFERENT MEDIUMS

✓ Discuss the effects of refraction with examples.

Activity 9.2: Explore the refraction of light in different mediums.

What I need:

(A transparent glass jar, cardboard, scissors, permanent marker and ruler) one set per group

What to do:

Draw an arrow (→) on the cardboard. Cut the cardboard into the arrow.

Measure the size of the arrow and note it down in the observation column (table 9.2).

Place the transparent glass jar on the table. Then, place the arrow behind the glass.

Observe the differences in the shape, position and size of the arrow. Now add water in the jar and record observations in table 9.2.

	Size of arrow	Position of arrow	Shape of arrow
Before placing behind the glass			
Behind the empty glass			
Behind the glass, filled with water			

Table 9.2: Observations for refraction of light

Activity Questions:

1. How does change in the speed of light on passing through a denser medium affect the direction of light?
2. How does the change in the speed and direction of light affect the image of the object present on the other side of the:
(A) Empty glass; and (b) Glass filled with water?

Teacher Note: Divide the students into groups.

Provide one set of materials to each group and ask them to perform the experiment and observe the changes that will take place in size, shape and position of arrow.

Real and Apparent Depth:

When a light ray travels from a rare medium into a denser medium it is refracted the ray bends towards the normal. However, when the same light ray travels from a denser to the rare medium its refracted ray moves away from the normal. When this refracted light enters human eye, it makes a shallower image of the object under water. Did you observe the difference in the real and apparent depth of the bed of a swimming pool or a lake? Even, fishermen experience the same issue of having a closer vision of fish in water than its real depth.

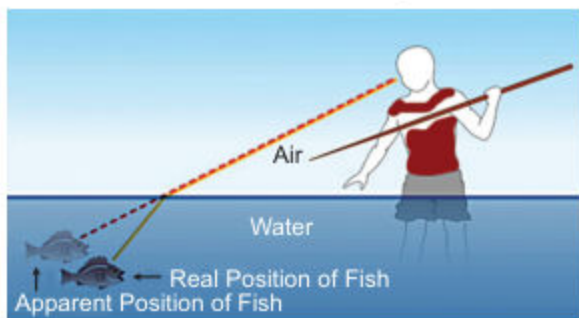


Figure 9.5: Real and apparent depth

Activity 9.3: Explore the real and apparent depth.

What I need:

Water, a wide-mouthed plastic bowl and a 5 rupee coin (one set per group).

What to do:

Take the coin and put it in the plastic bowl. Now, place the bowl on the table and one member of the group will position herself/himself in a way that coin should not be visible to her/him.



Figure 9.6: Visibility of coin and water level in the bowl

Hint

First member of the group will hold her/his position at the same height during the activity.

In the meanwhile, another group member will pour water in the bowl slowly and gradually.

Once the bowl is one-fourth filled, ask the observer if she/he can see the coin.

Pour more water to raise its level to the half of the bowl. Ask the observer if the coin is visible to her/him.

Finally, fill the bowl to its edges and ask the observer if the coin is visible to him/her.

Activity Questions	What I observed:
Is coin visible, when the bowl is one-fourth filled with water?	
Is coin visible, when the bowl is half filled?	
Is coin visible, when the bowl is filled to its edge?	
If coin is visible, give the reason for it.	

Critical Angle:

The angle of incidence, for which the angle of refraction is 90° , is known as the critical angle [as shown in figure 9.7 (b)]. It is denoted by 'c'. As discussed in the previous section, light rays passing through the rare medium from the denser, divert from the normal.

Consequently, the angle of refraction is greater than the angle of incidence. The greater is the difference between the angles of incidence and refraction, shallower will be the image of an underwater object [as shown in figure 9.7(a)]. If the angle of incidence is increased, there will be a time when maximum refraction occurs. As a result, the angle of refraction becomes 90° . In this situation, refracted light rays become parallel to the surface of the refracting medium.

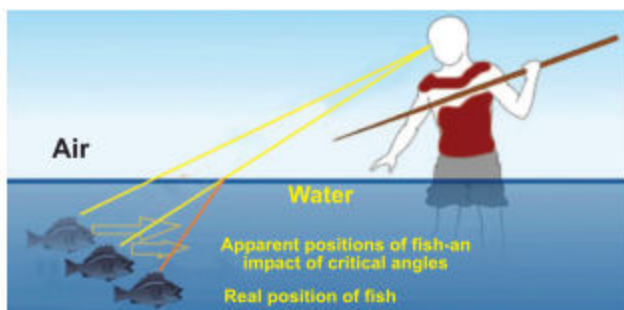


Figure 9.7 (a): Impact of Critical Angle

Table 9.3 shows critical angles of the different media.

Medium	Critical angle
Water	48.8°
Turpentine oil	44.1°
Crown glass	41.1°
Diamond	24.4°

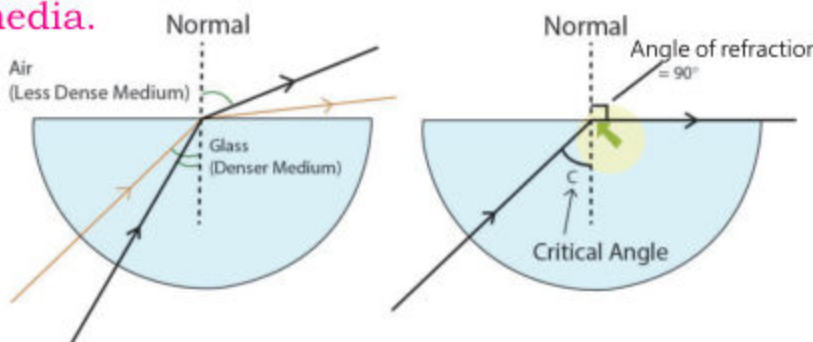


Figure 9.7 (b): Critical angle

Total Internal Reflection:

As we have studied, critical angle (c) is the angle of incidence for which angle of refraction is 90° . However, when the angle of incidence is greater than the critical angle ' c ', the light rays reflect back in the same denser medium. This phenomenon is named as the total internal reflection of the light rays as shown in figure 9.8 (a). There are two conditions for total internal reflection:

- Light rays travel from a denser to the rare medium.
- The angle of incidence of all the rays will be greater than the angle of their refraction.

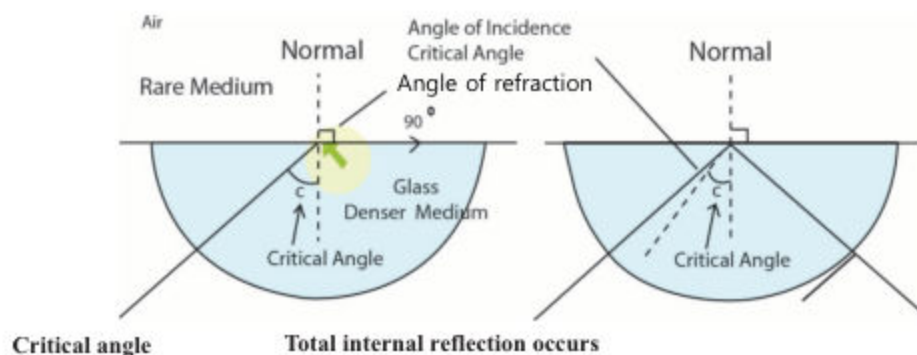


Figure 9.8 (a): Degree of C for total Internal reflection

Total internal reflection has many applications in our daily life:

(I) Reflecting prisms:

Reflecting prisms usually work on the principle of total internal reflection. Optical (visual) instruments like prismatic binocular use such prisms to invert, deviate and displace the light rays as shown in figure

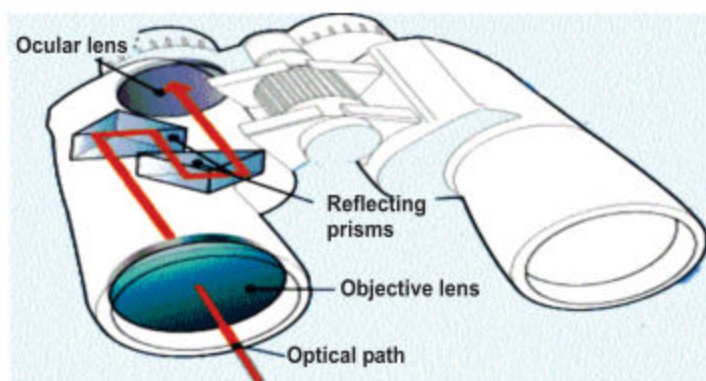


Figure 9.8 (b): Reflecting prisms

9.8 (b). These prisms deviate the incident rays coming from a distant object at 180° that reforms a clear and erect image of that object. Without these prisms, object's image will appear upside down to the user of the binocular.

(ii) Periscope: Periscope is another optical instrument that works on the principle of total internal reflection. It has two mirrors fitted into each end of the tube at an angle of 45° in a way that they face each other. When light rays hit the top mirror at 45° , they reflect away at the same angle. As a result, the light rays bend towards the bottom mirror and make the object visible to the observer, as shown in figure 9.8 (c).

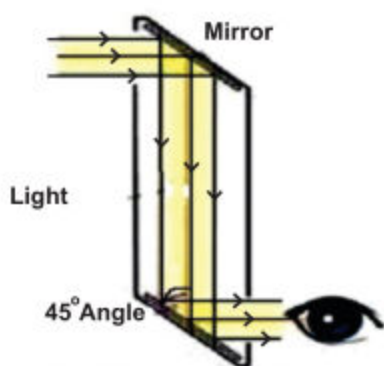


Figure 9.8 (c): Periscope

(iii) Mirage: Mirage is a phenomenon that occurs, in deserts and seas, due to which illusion of water appear away from their actual position. Mirage occurs from the refraction of light through two non-uniform mediums. Total internal reflection results in the mirage. On a hot day, a car driver may experience mirage as a puddle of water that appears on road several yards in front of his car.

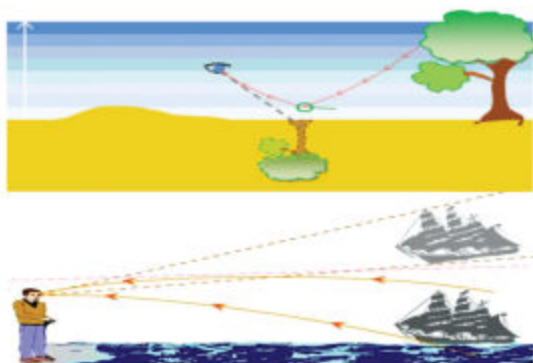


Figure 9.8 (d): Effects of total internal reflection

(iv) Fish Eye View: Fish and other aquatic animals see a reflected view of the objects, present in their surroundings as shown in figure 9.8 (e). When light reflects from the outer world to the boundary of two non-uniform media such as air and water, total internal reflection happens. As a result, outer world objects such as the moon, sun, plants and birds are visible to the fish under water. Simultaneously, other sea animals and plants present under water and even on the seabed are visible to each other. However, the shape and the distance of the image of such objects are different from their real frame.

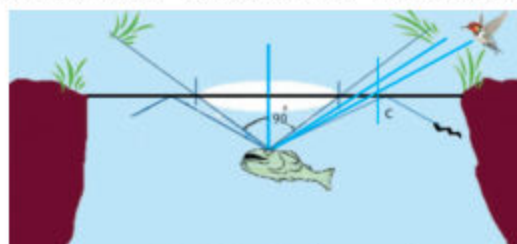


Figure 9.8 (e): Fish eye view- total internal reflection

DISPERSION OF LIGHT

- ✓ Describe the dispersion of light by using a prism.
- ✓ List colours of light using a prism. Define Spectrum.

The splitting of light into its constituent components is called dispersion of light. This phenomenon can be observed by using a dispersive prism.

The seven component colours of white light are red, orange, yellow, green, blue, indigo and violet.

DO YOU KNOW?

Triangular prism is one of the most commonly used dispersive prisms. Water droplets, water waves, dew drops and rock quartz crystals in air also act as dispersive prisms.

Activity 9.4: Demonstrate the dispersion of light by using a prism.

What I need:

A glass prism, a torch or a mobile phone light, drawing board or a table (to be used to set up the apparatus), scotch tape, a cardboard with slit in the centre, another rectangular cardboard to be used as the screen and a white sheet

What to do:

1. Wrap a white sheet around the rectangular cardboard (it will be used as the screen).
2. Set the apparatus as shown in figure 9.9.
3. Spot mobile phone or torch light on the slit of the cardboard.
4. Position the prism for wide and clear spectrum of seven colours.

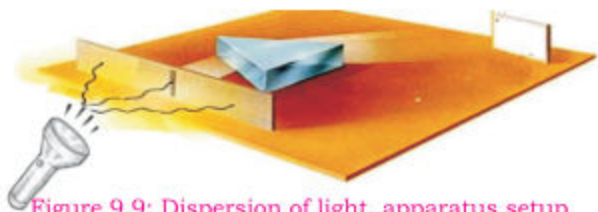


Figure 9.9: Dispersion of light, apparatus setup

Note the colour stream on the screen.

Activity Questions

How many colours can be observed on the screen?

What is the order of these colours, as they appear on screen?

What I observe:

Spectrum and Rainbow formation:

As we have discussed earlier, a band of seven colours: red, orange, yellow, green, blue, indigo and violet appear on the dispersion of white light. This band is called the spectrum of white light. If you could recall, monsoon season and that amazing multi-coloured arch appear in the sky after rain. That mesmerising multi coloured arch is known as rainbow the best example of the spectrum of white light in nature. You can identify the order of the colours of the rainbow as of the colours of the spectrum of white light, you observed in activity 9.4.



Figure 9.10 Coloured arch-the rainbow

Activity 9.5: Demonstrate how spinning of a rainbow (spectrum) results in the appearance of the white disc.

What I need:

A white cardboard, a pencil and eraser, round lid of a tin, ruler, scissors, crayons/colour pencils of (red, orange, yellow, green, blue, indigo and violet)

Demonstrate how spinning of a rainbow results in the appearance of white disc

What to do:

1. Place cardboard on the table. Use the round lid to trace a circle on the cardboard.
2. Cut out the circle with scissors.
3. Divide the circle into seven sections and colour them.
4. Make a hole in the centre of the coloured wheel.
5. Insert the pencil in the hole.
6. Spin the wheel at different speeds and record your observation.

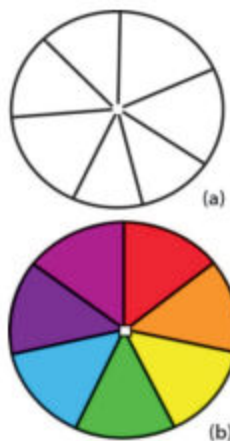


Figure 9.11: Colour wheel

Activity Questions	What I observe:
Can you see the colours, when you spin the wheel slowly?	Yes/No
If yes, then, which colour(s) could you see?	
Can you see the colours on changing the speed of wheel to its fullest?	Yes/No
If yes, then, which colour(s) could you see?	

COLOURS OF LIGHT

- ✓ Identify primary colours and show how they are combined to form secondary colours.

Although white light is composed of seven colours, still the red, green and blue are considered the primary colours of light. Can you justify the definition of primary colours?

The addition of primary colours in equal intensities results in the formation of secondary colours.

DO YOU KNOW?

Three colours of light that produce white light on combining with the correct intensity are called primary colours of light, such as red, green and blue.

Activity 9.6: Demonstrate how do the primary colours form secondary colours?

What I need:

Three colours lights (R=red, G=green and B=blue), white sheet, pencil and eraser

What to do:

Place a white sheet on the table and use the combination of any two lights as shown in figure 9.12.

Record your observations.

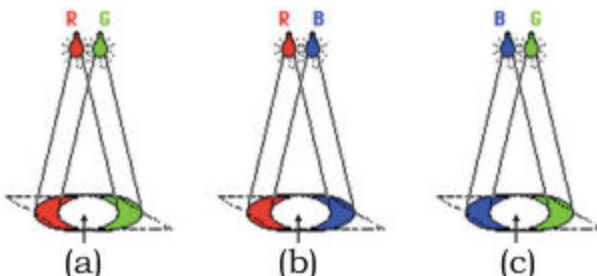
Activity Questions	What I observe?
Name the colour, appears on the spot (a), (b) and (c).	
Are the colours on (a), (b) and (c) spots primary or secondary? Answer:	
Give the reason for your answer, if they are primary or secondary colours :	

Figure 9.12

COLOURS OF OBJECTS

- ✓ Explain why an opaque or non-luminous object appears to be of certain colour.
- ✓ Identify different uses of light of different colours at home, school and country and explain the relationship of choice of colour to their purpose.
- ✓ Identify a device in their surroundings that uses different combinations of colours.

When white light strikes on the non-luminous objects, they reflect some of its colours while absorbing all the others. Consequently, that non-luminous object appears in the colour, it reflects. For example, rose looks red when it reflects red light while absorbs all the other colours of the white light. However, there are tens of other colours of roses which can be observed in Sindh. Each rose flower has distinctive colour depends on its ability to which colour of light it reflects. Can you name the other colours of roses cultivated in Sindh?

However, when an object reflects the entire component colours of light, it appears white. In contrast, any non-luminous object appears black when it absorbs all the component colours of light. Different colours of light are used for the different purposes in our daily life. Some uses of the colours of light are given below:

1. At Home:

- For home decoration, both the pastel and bright colours are used. The choice of the colour of walls, curtains and the furniture depends on the weather of the area. Residents of

Karachi and Hyderabad prefer pastel colours for their houses to create the cooling and relaxing effect in the prolonged summer season.

- (ii) In washrooms, toilets and other washing areas, red rims are used on the taps to show the warm water connection, while blue rims are on the taps to show the cold water supply.

2. At School:

- (i) Bright colours are used to build a kindergarten school unit because bright colours create excitement and interest for learning among the young children.
- (ii) Teachers use red or green colour for marking the students' assignments.
- (iii) Different colour markers and chalks are used to explain different topics on the white and black or green boards respectively.
- (iv) Colourful charts are used as the teaching aids to highlight different educational concepts.

3. At Country Level:

- (i) Traffic light signals use red, yellow and green lights to guide the drivers and pedestrians how to behave on roads. For instance, red traffic light instructs the drivers to stop; yellow light instructs them to be cautious, while green instruct them to keep driving.
- (ii) Red, yellow and orange lights and opaque objects are used to warn about the danger or cautionary actions on the public places. For instance, yellow strip indicates the edge of road for safe driving.
- (iii) In building construction, white and pastel colours are used to construct hospitals because these colours give a soothing effect to the environment.
- (iv) Ambulances and fire brigade use red and yellow light alarms to pass signals of emergency to other vehicle drivers.

- (v) Blue light is a standard treatment for jaundice among newborn babies.

Devices like the keypads of laptops use different coloured keys for different purposes. For instance, a blue colour dot illuminates to show the connectivity of laptop with internet or Wi-Fi. It turns red when internet or Wi-Fi signals are not available.

Activity 9.7: Identify the devices in the surroundings that use different combinations of colours.

What I need:

A pencil/pen

What to do:

Follow the teacher's instructions and act accordingly.
Make a list of devices that use different combinations of colours.

Activity Questions	What I observe:
Name the devices that use different combinations of colours.	

Teacher Note: Divide students into groups. Ask each member of the group to observe their homes, neighbourhood, school premises and close by markets for 2 days during day and evening hours.

Ask them to look for the devices that use different combinations of colours and give at least two examples.

Summary

- When light passes from a rare to the denser medium, it changes its speed and bends, this bending of light is called refraction of light.
- Refraction of light helps to form images in our eyes and assists water animals to see from inside to outside of the water. Rainbow is formed due to refraction of light.
- Refractive index is the ratio between the speed of light in vacuum and its speed in any other medium. It is a constant and can be calculated as:
- Refractive Index (n) of a medium =
$$\frac{\text{Speed of light in vacuum (c)}}{\text{Speed of light in the medium (v)}}$$
- Critical angle (c) is the angle of incidence for which angle of refraction is 90° .
- Total internal reflection of the light occurs when the angle of the incidence of the refracted light rays is greater than the critical angle ' c '.
- Mirage and fish eye view are the two applications of the total internal reflection of light.
- Binocular and periscope also work on the principle of total internal reflection of light.
- Light on passing through a prism refracts and breaks up into its seven component colours: red, orange, yellow, green, blue, indigo and violet. This phenomenon is known as the dispersion of light.
- The band of seven colours is called spectrum.
- Rainbow is the natural spectrum, formed after the rain when droplets of water in the air serve as the prism and break white sunlight into seven colours.
- Red, blue and green are called primary colours of light. These primary colours combine in equal intensities to form the secondary colours of light.
- The colour of an opaque or non-luminous object is the colour of light it reflects. For instance, grass seems green because it reflects only green light out of the seven colours of white light.
- White objects reflect all the seven colours of light while black objects absorb all the seven colours of light.

Review Exercises

1. Fill in the blanks:
 - i) When light enters a right-angled prism, it makes an angle _____ than the critical angle.
 - ii) Splitting of white light into its component colours is called _____.
 - iii) Critical angle of water is _____.
 - iv) A binocular uses _____ prisms to see distant objects.
 - v) If an object absorbs all seven colours of light, it appears in _____ colour.
2. Describe five uses of different colours of light other than the ones stated in the chapter.
3. Draw the labelled diagrams of refraction in water and glass.
4. Compare the following:
 - i) Real and apparent depth
 - ii) Primary and secondary colours
5. (a) State the laws of refraction.
(b) Discuss refractive index with the help of two examples.
6. Explain the following phenomena with the help of their labelled diagrams:
 - i) Mirage
 - ii) Fish eye view

PROJECT

Different phenomena of light such as reflection, refraction and dispersion have different significance in our life. These phenomena play a crucial role in the lives of humans and other animals. We can see the colours of nature and enjoy many devices that work on the principles of these three phenomena. Investigate your surroundings for a week and collect the evidence of refraction, total internal reflection and dispersion of light.

- (a)** Make a list of the events in which you observed refraction, total internal reflection and the dispersion of light.
- (b)** Find out the factors that contributed in the refraction, total internal reflection and the dispersion of light.
- (c)** Mention at least one incidence in which reflection, refraction and total internal reflection occur simultaneously.
- (d)** What will happen to our life, if refraction and dispersion of light do not happen in nature?